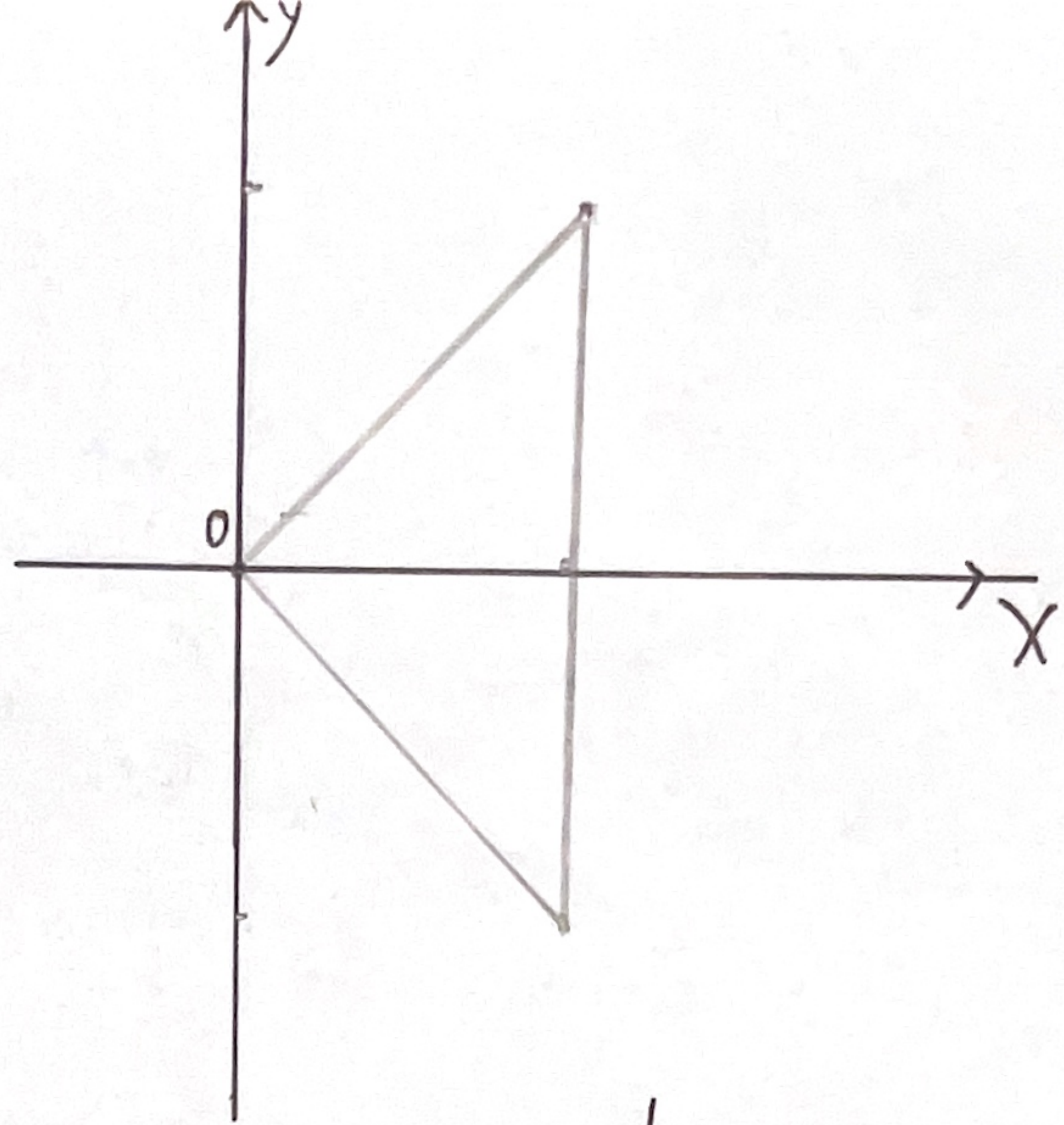
六、



解：(1)

先求边缘缘概率密度

求 $f_Y(y)$

① 当 $0\le y\le 1$

$$f_Y(y)=\int_y^1 f(x,y)\,dx=\int_y^1 1\cdot dx=1-y$$

② 当 $-1<y\le 0$

$$f_Y(y)=\int_{-y}^1 f(x,y)\,dx=\int_{-y}^1 1\cdot dx=1+y$$

∴ 当 $-1<y\le 1$

$$f_Y(y)=1-|y|$$

求 $f_X(x)$

$$f_X(x)=\int_{-x}^{x} 1\cdot dy=2x$$

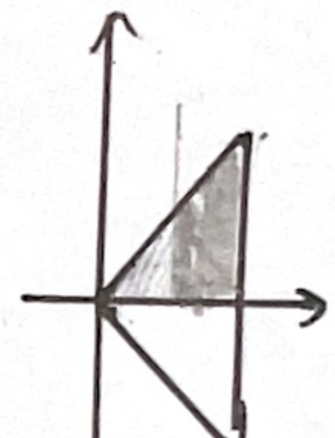
求 $f_{X|Y}(x|y)$

$$f_{X|Y}(x|y)=\frac{f(x,y)}{f_Y(y)}=\frac{1}{1-|y|}$$

$$f_{X|Y}(x|y)=\begin{cases}\dfrac{1}{1-|y|} & |y|<x\text{ 且 }0<x<1\\ 0 & \text{其它}\end{cases}$$

同理 $f_{Y|X}(y|x)=\begin{cases}\dfrac{1}{2x} & |y|<x\text{ 且 }0<x<1\\ 0 & \text{其它}\end{cases}$

(2)



$$P(X>\tfrac{1}{2}|Y>0)=\frac{P(X>\frac{1}{2},Y>0)}{P(Y>0)}$$

$$=\frac{\int_{\frac{1}{2}}^1 dx\int_0^x f(x,y)dy}{\int_0^1 f_Y(y)dy}\quad\text{or}\quad=\frac{\int_{\frac{1}{2}}^1 dx\int_0^x f(x,y)dy}{\int_0^1 dx\int_0^x f(x,y)\,dy}$$

对应区域

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$$=\frac{\int_{\frac{1}{2}}^1 dx\int_0^x dy}{\int_0^1(1-y)dy}=\frac{\frac{1}{2}\cdot\frac{3}{4}}{\frac{1}{2}}=\frac{3}{4}$$

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(3) ∵ $f_{X|Y}(x|y)=\dfrac{1}{1-|y|}$

∴ $f_{X|Y}(x|\frac{1}{2})=\dfrac{1}{1-1/2}=2$

∵ $f_{X|Y}(x|\frac{1}{2})$ 为在 $Y=1/2$ 的条件下随机变量 X 的条件概率密度

∴ $P(X>\frac{2}{3}|Y=\frac{1}{2})=\int_{\frac{2}{3}}^{+\infty} f_{X|Y}(x|\frac{1}{2})\,dx$

$=\int_{\frac{2}{3}}^1 f_{X|Y}(x|\frac{1}{2})\,dx=\int_{\frac{2}{3}}^1 2\,dx=\frac{2}{3}$